



Department of Mathematics  
IIT Hyderabad  
MA1110: Calculus I  
Credits: 1  
Fall 2026

*Syllabus & Evaluation*

Instructor: Prof. Prasun Roychowdhury

Office: MA 414

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**Syllabus:**

**Sequences and Series:** Basic properties of the real numbers, Limit of a sequence, monotone and Cauchy sequences, and properties of convergent sequences, examples. Infinite series, positive series, tests for convergence and divergence, integral test, alternating series, Leibniz test.

**Differential Calculus:** Continuity and differentiability of a function of a single variable, statement of Rolle's Theorem, Lagrange's Mean Value Theorem, and applications.

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**Suggested Texts:**

1. **Maurice Weir, and Joel Hass:** *Thomas' Calculus: Early Transcendentals*, Pearson, 14th Edition, 2018.
2. **James Stewart:** *Calculus: Early Transcendentals*, Cengage India Private Limited, 7th Edition, 2017.
3. **Sadhan Kumar Mapa:** *Intoduction to Real Analysis*, Sarat Book Distributors, 6th Edition, 2006.
4. **Tom M. Apostol:** *Mathematical Analysis*, Second edition, 1973.
5. **Walter Rudin:** *Principles of Mathematical Analysis*, McGraw Hill Education, Third edition, 1976.

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**Prerequisites:**

Elementary algebra, set theory, functions, trigonometry, and elementary analytic geometry.

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**Aims:**

The course aims to provide a rigorous introduction to the fundamental concepts of single-variable calculus. Emphasis is placed on the study of sequences and infinite series, continuity, differentiability, and the fundamental theorems of differential calculus. The course aims to develop mathematical reasoning, analytical thinking, and problem-solving skills, while laying a foundation for subsequent studies in mathematics, science, and engineering.

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**Office Hours:**

By appointment. Students may schedule a meeting by email at [prasunrc@math.iith.ac.in](mailto:prasunrc@math.iith.ac.in).

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**Practice Problems:**

Two sets of practice problems will be posted on the course [webpage](#) during the course. These are for practice and will not be evaluated unless announced otherwise.

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**Evaluation Scheme:**

A single comprehensive examination covering all topics of the course will be conducted at the end of the course. The tentative date of the examination is **29 August 2026**. The examination will consist of Multiple Select Questions (MSQs). The examination date and question format are expected to be as stated above; however, any changes, if necessary, will be announced in due course.

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